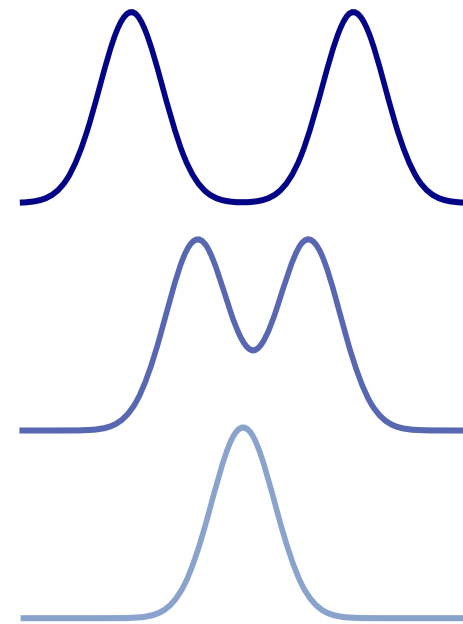




► Parallel tempering constructs MCMC chain $\mathbf{X}_t$ on $\mathbb{X}^{N+1}$ targetting $\pi_{\beta_0} \otimes \cdots \otimes \pi_{\beta_N}$

$$\mathbf{X}_t = \begin{pmatrix} X_t^N \\ \vdots \\ X_t^n \\ \vdots \\ X_t^0 \end{pmatrix} \quad \begin{matrix} \sim \pi_{\beta_N} \\ \\ \sim \pi_{\beta_n} \\ \\ \sim \pi_{\beta_0} \end{matrix}$$

Target States
Interpolating States
Reference States





PARALLEL TEMPERING

- Generate $\mathbf{X}_t$ from $\mathbf{X}_{t-1}$ using by applying the local exploration and communication moves

$$\mathbf{X}_t \sim \mathbf{K}_{\text{PT}}(\mathbf{X}_{t-1}, d\mathbf{x}') \quad \mathbf{K}_{\text{PT}} = \mathbf{K}_{\text{expl}}\mathbf{K}_{\text{comm}}$$



Independently discovered in statistics and physics

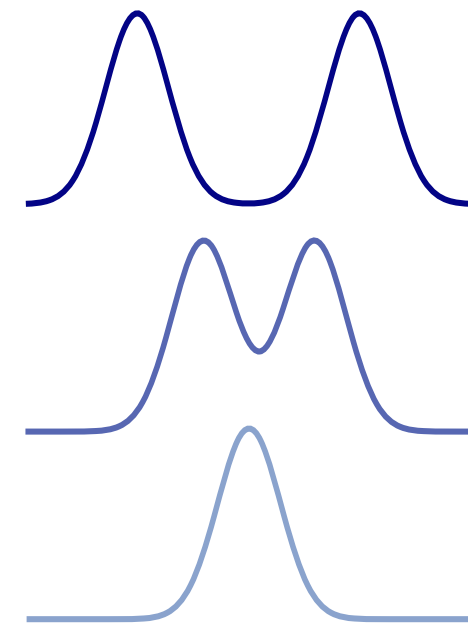


~~Koji Hukushima and Koji Nemoto in 1995 in physics~~

PARALLEL TEMPERING

- ▶ Independently discovered in statistics and physics
 - ▶ Charles Geyer in 1991 in statistics
 - ▶ Koji Hukushima and Koji Nemoto in 1995 in physics
- ▶ Parallel tempering constructs MCMC chain $\mathbf{X}_t$ on $\mathbb{X}^{N+1}$ targetting $\pi_{\beta_0} \otimes \cdots \otimes \pi_{\beta_N}$

$$\mathbf{X}_t = \begin{pmatrix} X_t^N \\ \vdots \\ X_t^n \\ \vdots \\ X_t^0 \end{pmatrix} \quad \begin{array}{ll} \sim \pi_{\beta_N} & \text{Target States} \\ \sim \pi_{\beta_n} & \text{Interpolating States} \\ \sim \pi_{\beta_0} & \text{Reference States} \end{array}$$



- ▶ Generate $\mathbf{X}_t$ from $\mathbf{X}_{t-1}$ using by applying the local exploration and communication moves

$$\mathbf{X}_t \sim \mathbf{K}_{\text{PT}}(\mathbf{X}_{t-1}, d\mathbf{x}') \quad \mathbf{K}_{\text{PT}} = \mathbf{K}_{\text{expl}} \mathbf{K}_{\text{comm}}$$

LOCAL EXPLORATION